

CrowdStrike Falcon + Wireshark + Sysmon True Positive Investigation

Suspected C2 beacons confirmed by packet and endpoint evidence

TRUE POSITIVE

Confirmed incident with process-linked C2

Independent evidence review across network, packet, endpoint, and host telemetry.
Complete source evidence is preserved at full size on the following 15 pages.



TRUE POSITIVE

Investigation
overview

EVIDENCE TRACK

- CASE BRIEF
- ALERT + RAW EVIDENCE
- PACKET VERIFICATION
- ENDPOINT CORROBORATION
- CORRELATION + JUDGMENT
- DETECTION ENGINEERING
- DISPOSITION + COMMUNICATION

Professional CrowdStrike Falcon + Wireshark + Sysmon

True Positive Investigation

Suspected C2 Beaconsing Alert Confirmed by Packet and Endpoint Evidence

One high-confidence portfolio scenario showing how a SIEM C2 alert is validated as a true positive when the raw network-device logs, Wireshark packet capture, CrowdStrike Falcon endpoint telemetry, and Sysmon evidence independently corroborate the same malicious activity chain.

CORE FINDING

Falcon Next-Gen SIEM reported periodic outbound command-and-control beaconsing from FIN-WS-023 to 203.0.113.77. The raw firewall records confirmed allowed outbound sessions with non-zero bytes. Wireshark captured the same repeated TCP/TLS sessions. Falcon tied all twelve connections to an unknown user-space process, healthsync.exe, launched from suspicious PowerShell lineage. Sysmon independently recorded the process, DNS lookup, twelve network connections, file creation, and Run-key persistence. The alert is therefore a confirmed TRUE POSITIVE.

Investigation Workflow

1. SIEM Alert	2. Raw Source Validation	3. Wireshark	4. Falcon EDR	5. Sysmon	6. Disposition
High-severity C2 beaconsing	Outbound + allowed + bytes transferred	12 matching outbound TLS sessions	12 process-linked connections to IOC	Process + DNS + network + file + Run-key evidence	TRUE POSITIVE + containment

Professional Use Notes

Note	Professional Handling
Sanitized indicators	All hosts, users, IPs, domains, identifiers, hashes, and logs are synthetic portfolio data. Documentation IP ranges and the .invalid TLD are used for examples.
Packet evidence	Wireshark is treated as independent packet-level corroboration because the capture interface, time window, clock alignment, and packet-drop status are documented.
Endpoint corroboration	The true-positive decision does not rely on one alert or IOC. Raw firewall logs, packet capture, Falcon process/network telemetry, and Sysmon independently converge on the same chain.
True-positive definition	The SIEM asserts outbound C2, and independent sources prove that the endpoint initiated allowed sessions, a specific process generated them, and malicious state changes occurred on the host.
Safe lab ground truth	healthsync.exe, sync-check.example.invalid, and 203.0.113.77 are synthetic adversary-emulation indicators. No live malware or third-party target is required.

Alert intake and trigger evidence

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1. SIEM Alert Intake and Raw Trigger Evidence

At 12:32:14Z, CrowdStrike Falcon Next-Gen SIEM raises a High alert titled Possible C2 Beaconing - Repetitive HTTPS Sessions. The entity FIN-WS-023 is associated with internal address 10.40.23.55, and the alert identifies twelve outbound connections to 203.0.113.77:443 at approximately 60-second intervals.

ANALYST PRINCIPLE

Treat the alert as a hypothesis until the raw source and endpoint evidence are reviewed. A true positive is established only when the underlying network direction, session disposition, packet sequence, process identity, and endpoint effects corroborate the alert.

Alert Summary

Field	Value	Analyst Focus
Alert time	2026-08-18T12:32:14.901Z	High-severity correlation fired after 12 repeated sessions.
SIEM	CrowdStrike Falcon Next-Gen SIEM	Correlation output must be validated against raw telemetry.
Alert title	Possible C2 Beaconing - Repetitive HTTPS Sessions	Narrative asserts outbound command-and-control behavior.
Entity	FIN-WS-023 / 10.40.23.55	Host attribution must match the traffic initiator.
Remote IP	203.0.113.77:443	Synthetic external C2 indicator for the scenario.
Remote domain	sync-check.example.invalid	Synthetic controlled lab domain.
Observed count	12 sessions	Repeated connections are sufficient to calculate periodicity.
Mean interval	60.0 seconds	Stable cadence is strongly consistent with beaconing when sessions are established.
Process	Unknown in firewall-only correlation	Endpoint telemetry is required to identify the responsible process.
Initial severity	High	Escalate for packet and endpoint corroboration.

Raw Firewall Records That Fed the SIEM

Line	Raw Source	Raw SIEM / Firewall Log	Analyst Read
01	firewall_raw	2026-08-18T12:20:02Z FW-EDGE-01 vendor=ExampleFW direction=outbound src=10.40.23.55 spt=52110 dst=203.0.113.77 dpt=443 proto=tcp state=established action=allow rule=EGRESS-WEB sent_bytes=760 rcv_bytes=612 session_bytes=1372	Allowed outbound established TLS session with non-zero data transfer.
02	firewall_raw	2026-08-18T12:21:02Z FW-EDGE-01 vendor=ExampleFW direction=outbound src=10.40.23.55 spt=52147 dst=203.0.113.77 dpt=443 proto=tcp state=established action=allow rule=EGRESS-WEB sent_bytes=779 rcv_bytes=629 session_bytes=1408	Allowed outbound established TLS session with non-zero data transfer.
03	firewall_raw	2026-08-18T12:22:02Z FW-EDGE-01 vendor=ExampleFW direction=outbound src=10.40.23.55 spt=52184 dst=203.0.113.77 dpt=443 proto=tcp state=established action=allow rule=EGRESS-WEB sent_bytes=798 rcv_bytes=646 session_bytes=1444	Allowed outbound established TLS session with non-zero data transfer.
04	firewall_raw	2026-08-18T12:23:02Z FW-EDGE-01 vendor=ExampleFW direction=outbound src=10.40.23.55 spt=52221 dst=203.0.113.77 dpt=443 proto=tcp state=established action=allow rule=EGRESS-WEB sent_bytes=817 rcv_bytes=663 session_bytes=1480	Allowed outbound established TLS session with non-zero data transfer.
05	firewall_raw	2026-08-18T12:24:02Z FW-EDGE-01 vendor=ExampleFW direction=outbound src=10.40.23.55 spt=52258 dst=203.0.113.77 dpt=443 proto=tcp state=established action=allow rule=EGRESS-WEB sent_bytes=760 rcv_bytes=680 session_bytes=1440	Allowed outbound established TLS session with non-zero data transfer.

Raw records and source validation

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Raw Firewall Records - Continued and Source Validation

Line	Raw Source	Raw SIEM / Firewall Log	Analyst Read
06	firewall_raw	2026-08-18T12:25:02Z FW-EDGE-01 vendor=ExampleFW direction=outbound src=10.40.23.55 spt=52295 dst=203.0.113.77 dpt=443 proto=tcp state=established action=allow rule=EGRESS-WEB sent_bytes=779 recv_bytes=612 session_bytes=1391	Allowed outbound established TLS session with non-zero data transfer.
07	firewall_raw	2026-08-18T12:26:02Z FW-EDGE-01 vendor=ExampleFW direction=outbound src=10.40.23.55 spt=52332 dst=203.0.113.77 dpt=443 proto=tcp state=established action=allow rule=EGRESS-WEB sent_bytes=798 recv_bytes=629 session_bytes=1427	Allowed outbound established TLS session with non-zero data transfer.
08	firewall_raw	2026-08-18T12:27:02Z FW-EDGE-01 vendor=ExampleFW direction=outbound src=10.40.23.55 spt=52369 dst=203.0.113.77 dpt=443 proto=tcp state=established action=allow rule=EGRESS-WEB sent_bytes=817 recv_bytes=646 session_bytes=1463	Allowed outbound established TLS session with non-zero data transfer.
09	firewall_raw	2026-08-18T12:28:02Z FW-EDGE-01 vendor=ExampleFW direction=outbound src=10.40.23.55 spt=52406 dst=203.0.113.77 dpt=443 proto=tcp state=established action=allow rule=EGRESS-WEB sent_bytes=760 recv_bytes=663 session_bytes=1423	Allowed outbound established TLS session with non-zero data transfer.
10	firewall_raw	2026-08-18T12:29:02Z FW-EDGE-01 vendor=ExampleFW direction=outbound src=10.40.23.55 spt=52443 dst=203.0.113.77 dpt=443 proto=tcp state=established action=allow rule=EGRESS-WEB sent_bytes=779 recv_bytes=680 session_bytes=1459	Allowed outbound established TLS session with non-zero data transfer.
11	firewall_raw	2026-08-18T12:30:02Z FW-EDGE-01 vendor=ExampleFW direction=outbound src=10.40.23.55 spt=52480 dst=203.0.113.77 dpt=443 proto=tcp state=established action=allow rule=EGRESS-WEB sent_bytes=798 recv_bytes=612 session_bytes=1410	Allowed outbound established TLS session with non-zero data transfer.
12	firewall_raw	2026-08-18T12:31:02Z FW-EDGE-01 vendor=ExampleFW direction=outbound src=10.40.23.55 spt=52517 dst=203.0.113.77 dpt=443 proto=tcp state=established action=allow rule=EGRESS-WEB sent_bytes=817 recv_bytes=629 session_bytes=1446	Allowed outbound established TLS session with non-zero data transfer.
Line	Raw Source	Raw SIEM / Normalization Log	Analyst Read
13	parser_raw	2026-08-18T12:32:10.202Z Parser=custom_firewall_v4 InputDirection=outbound InputAction=allow OutputLocalIP=10.40.23.55 OutputRemoteIP=203.0.113.77 OutputDirection=outbound OutputAction=allow OutputBytes=1630	Parser preserves direction, action, and byte evidence.
14	asset_enrichment	2026-08-18T12:32:10.219Z LocalIP=10.40.23.55 Asset=FIN-WS-023 Confidence=high Source=CMDB Role=Finance-Workstation	Source/local initiator correctly maps to the endpoint entity.
15	correlation_raw	2026-08-18T12:32:14.880Z Rule=Periodic_External_HTTPS GroupBy=Asset,RemoteIP Count=12 AvgInterval=60.0s Threshold=8 DirectionConstraint=outbound ActionConstraint=allow SessionEvidence=established	Rule evaluates established outbound allowed sessions.
16	alert_raw	2026-08-18T12:32:14.901Z Alert=Possible C2 Beaconing - Repetitive HTTPS Sessions Severity=High Host=FIN-WS-023 LocalIP=10.40.23.55 RemoteIP=203.0.113.77 RemotePort=443 Count=12 Process=unknown	High alert is consistent with the normalized network evidence.

Raw Source Validation Result

Validation	Observed Result	Meaning
Direction / initiator	outbound in all 12 source records; local/source IP=10.40.23.55	FIN-WS-023 is the connection initiator.
Disposition / state	action=allow and state=established in all 12 records	Traffic completed rather than being blocked or reset-only.
Bytes	non-zero sent and received bytes in every session	Bidirectional application data was exchanged.
Parser + periodicity	direction/action/bytes preserved; 12 sessions with mean interval 60.0 seconds	No normalization defect explains the alert; cadence supports automated callback behavior.

TRUE POSITIVE

Packet capture validation

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2. Wireshark Packet Verification - Endpoint Capture

Wireshark is opened on FIN-WS-023 and the full capture covering the alert window is preserved. If the SIEM narrative is correct, the packet capture should contain outbound SYNs, completed handshakes, TLS Client Hello packets, recurring encrypted application data, and matching destination timing.

Capture Integrity and Scope

Item	Recorded Value	Why It Matters
Capture host	FIN-WS-023 / 10.40.23.55	Matches the SIEM entity.
Interface	Ethernet 3 - active corporate NIC	Confirms the active network path was captured.
Capture window	2026-08-18T12:18:00Z - 12:36:00Z	Begins before initial execution and ends after the alert.
Packets captured	71,988	Continuous traffic provides sufficient visibility.
Capture drops	0 reported	Supports trustworthy packet correlation.
NTP drift	< 5 ms against domain time source	Allows direct timestamp comparison with Falcon/Sysmon/firewall logs.
Host IP observed	10.40.23.55	Confirms correct endpoint capture.
Target IOC filter	ip.addr == 203.0.113.77	Returns 56 frames.
Outbound SYN filter	ip.src == 10.40.23.55 && ip.dst == 203.0.113.77 && tcp.flags.syn == 1 && tcp.flags.ack == 0	Returns 12 frames.
TLS Client Hello filter	tls.handshake.type == 1 && ip.dst == 203.0.113.77	Returns 12 frames.

Raw Wireshark Packet Logs - Initial Beacon Establishment

Line	Frame	Raw Wireshark Packet Log	Packet Interpretation
01	1188	2026-08-18T12:19:59.884Z Src=10.40.23.55 Dst=10.40.0.10 Proto=DNS Info=Query A sync-check.example.invalid	Endpoint resolves synthetic C2 domain.
02	1189	2026-08-18T12:19:59.887Z Src=10.40.0.10 Dst=10.40.23.55 Proto=DNS Info=Response A sync-check.example.invalid=203.0.113.77	DNS returns the destination used by the beacon.
03	1210	2026-08-18T12:20:02.004Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TCP Info=52110 > 443 SYN stream=44	FIN-WS-023 initiates the first TCP session.
04	1211	2026-08-18T12:20:02.021Z Src=203.0.113.77 Dst=10.40.23.55 Proto=TCP Info=443 > 52110 SYN,ACK stream=44	Remote endpoint accepts the connection.
05	1212	2026-08-18T12:20:02.027Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TCP Info=52110 > 443 ACK stream=44	TCP handshake completes.
06	1214	2026-08-18T12:20:02.041Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TLShv1.3 Info=Client Hello SNI=sync-check.example.invalid ALPN=h2 stream=44	Endpoint begins encrypted application session.
07	1217	2026-08-18T12:20:02.073Z Src=203.0.113.77 Dst=10.40.23.55 Proto=TLShv1.3 Info=Server Hello stream=44	TLS negotiation continues.
08	1224	2026-08-18T12:20:02.105Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TLShv1.3 Info=Application Data len=384 stream=44	Client sends encrypted application data.
09	1227	2026-08-18T12:20:02.132Z Src=203.0.113.77 Dst=10.40.23.55 Proto=TLShv1.3 Info=Application Data len=196 stream=44	Remote side returns encrypted data.
10	1232	2026-08-18T12:20:02.161Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TCP Info=FIN,ACK stream=44	First callback closes cleanly.
11	1512	2026-08-18T12:21:02.006Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TCP Info=52147 > 443 SYN stream=45	Second callback begins approximately 60 seconds later.
12	1518	2026-08-18T12:21:02.043Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TLShv1.3 Info=Client Hello SNI=sync-check.example.invalid stream=45	Second TLS callback to same domain/IP.

TRUE POSITIVE

Packet evidence and targeted filters

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Wireshark Packet Logs - Continued

Line	Frame	Raw Wireshark Packet Log	Packet Interpretation
13	1829	2026-08-18T12:22:02.049Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TLShv1.3 Info=Client Hello SNI=sync-check.example.invalid stream=46	Third TLS callback.
14	1838	2026-08-18T12:22:02.109Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TLShv1.3 Info=Application Data len=392 stream=46	Third outbound encrypted message.
15	2140	2026-08-18T12:23:02.009Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TCP Info=52221 > 443 SYN stream=47	Fourth callback.
16	2451	2026-08-18T12:24:02.012Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TCP Info=52258 > 443 SYN stream=48	Fifth callback.
17	2764	2026-08-18T12:25:02.010Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TCP Info=52295 > 443 SYN stream=49	Sixth callback.
18	3077	2026-08-18T12:26:02.008Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TCP Info=52332 > 443 SYN stream=50	Seventh callback.
19	3391	2026-08-18T12:27:02.014Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TCP Info=52369 > 443 SYN stream=51	Eighth callback.
20	3708	2026-08-18T12:28:02.006Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TCP Info=52406 > 443 SYN stream=52	Ninth callback.
21	4022	2026-08-18T12:29:02.012Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TCP Info=52443 > 443 SYN stream=53	Tenth callback.
22	4339	2026-08-18T12:30:02.009Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TCP Info=52480 > 443 SYN stream=54	Eleventh callback.
23	4654	2026-08-18T12:31:02.013Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TCP Info=52517 > 443 SYN stream=55	Twelfth callback.
24	4661	2026-08-18T12:31:02.051Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TLShv1.3 Info=Client Hello SNI=sync-check.example.invalid stream=55	Twelfth TLS session.
25	4668	2026-08-18T12:31:02.108Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TLShv1.3 Info=Application Data len=416 stream=55	Twelfth outbound encrypted message.
26	4671	2026-08-18T12:31:02.134Z Src=203.0.113.77 Dst=10.40.23.55 Proto=TLShv1.3 Info=Application Data len=205 stream=55	Twelfth response.
27	4677	2026-08-18T12:31:02.165Z Src=10.40.23.55 Dst=203.0.113.77 Proto=TCP Info=FIN,ACK stream=55	Final observed callback closes.

Targeted Wireshark Validation Filters

ip.addr == 203.0.113.77 ip.src == 10.40.23.55 && ip.dst == 203.0.113.77 tcp.flags.syn == 1 && tcp.flags.ack == 0 && ip.src == 10.40.23.55 && ip.dst == 203.0.113.77 && tcp.dstport == 443 tls.handshake.type == 1 && ip.dst == 203.0.113.77 tls.handshake.extensions_server_name == "sync-check.example.invalid" dns.qry.name == "sync-check.example.invalid"		
Filter	Result	Interpretation
ip.addr == 203.0.113.77	56 frames	IOC repeatedly appears on the endpoint capture.
Outbound FIN-WS-023 -> IOC	29 frames	Endpoint sends traffic to the IOC across the alert window.
Outbound SYN to IOC:443	12 frames	Twelve distinct TCP sessions are initiated.
TLS Client Hello to IOC	12 frames	Every periodic connection negotiates TLS.
SNI sync-check.example.invalid	12 frames	All beacon sessions identify the same synthetic C2 domain.
DNS query for sync-check.example.invalid	1 query + response	Endpoint resolves the domain to 203.0.113.77 before beaconing.

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3. CrowdStrike Falcon Endpoint Telemetry - Process-Linked C2 Confirmed

The Falcon sensor is queried independently of the firewall correlation. ProcessRollup2 is used to reconstruct execution lineage and NetworkConnectIP4 is used to attribute the outbound connections to the responsible endpoint process.

Raw Falcon Endpoint Events During the Alert Window

Line	Falcon Event	Raw Falcon Event / Query Result	Analyst Read
01	ProcessRollup2	2026-08-18T12:19:41.502Z ComputerName=FIN-WS-023 aid=tp001 ImageFileName=C:\Program Files\Microsoft Office\root\Office16\WINWORD.EXE ParentBaseFileName=explorer.exe CommandLine=WINWORD.EXE C:\Users\mcarter\Downloads\QuarterlyReview.docm TargetProcessId=7412	Document opens shortly before suspicious execution.
02	ProcessRollup2	2026-08-18T12:19:43.114Z ComputerName=FIN-WS-023 aid=tp001 ImageFileName=C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe ParentBaseFileName=WINWORD.EXE CommandLine=powershell.exe -NoProfile -WindowStyle Hidden -EncodedCommand <redacted> TargetProcessId=8356	Suspicious PowerShell launches from Office.
03	ProcessRollup2	2026-08-18T12:19:45.082Z ComputerName=FIN-WS-023 aid=tp001 ImageFileName=C:\Users\mcarter\AppData\Roaming\HealthSync\healthsync.exe ParentBaseFileName=powershell.exe CommandLine=healthsync.exe --background TargetProcessId=9820 SHA256=9f0c7c1e...synthetic	Unknown user-space process launches from suspicious PowerShell.
04	NetworkConnectIP4	2026-08-18T12:20:02.004Z ComputerName=FIN-WS-023 aid=tp001 ContextBaseFileName=healthsync.exe ContextProcessId=9820 LocalAddressIP4=10.40.23.55 LocalPort=52110 RemoteAddressIP4=203.0.113.77 RemotePort=443	Falcon attributes first C2 connection to healthsync.exe.
05	NetworkConnectIP4	2026-08-18T12:21:02.006Z ComputerName=FIN-WS-023 aid=tp001 ContextBaseFileName=healthsync.exe ContextProcessId=9820 LocalAddressIP4=10.40.23.55 LocalPort=52147 RemoteAddressIP4=203.0.113.77 RemotePort=443	Second process-linked callback.
06	NetworkConnectIP4	2026-08-18T12:25:02.010Z ComputerName=FIN-WS-023 aid=tp001 ContextBaseFileName=healthsync.exe ContextProcessId=9820 LocalAddressIP4=10.40.23.55 LocalPort=52295 RemoteAddressIP4=203.0.113.77 RemotePort=443	Sixth callback sample.
07	NetworkConnectIP4	2026-08-18T12:29:02.012Z ComputerName=FIN-WS-023 aid=tp001 ContextBaseFileName=healthsync.exe ContextProcessId=9820 LocalAddressIP4=10.40.23.55 LocalPort=52443 RemoteAddressIP4=203.0.113.77 RemotePort=443	Tenth callback sample.
08	NetworkConnectIP4	2026-08-18T12:31:02.013Z ComputerName=FIN-WS-023 aid=tp001 ContextBaseFileName=healthsync.exe ContextProcessId=9820 LocalAddressIP4=10.40.23.55 LocalPort=52517 RemoteAddressIP4=203.0.113.77 RemotePort=443	Twelfth process-linked callback.
09	hunt_result	2026-08-18T12:35:10.100Z Query=NetworkConnectIP4 ComputerName=FIN-WS-023 RemoteAddressIP4=203.0.113.77 ContextProcessId=9820 ResultCount=12	Hunt confirms twelve Falcon endpoint connections to IOC.
10	sensor_health	2026-08-18T12:35:31.008Z ComputerName=FIN-WS-023 aid=tp001 SensorState=online EventFlow=normal	Sensor was online and collecting normally during the window.

Falcon pivots and result matrix

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Falcon Validation Pivots and Result Matrix

The process/network relationship is the critical Falcon contribution. The external destination is not merely associated with the host; it is linked to a specific executable and process ID that originated from suspicious parent-child execution.

Falcon CQL - Confirm Endpoint Connections to IOC

```
#event_simpleName=NetworkConnectIP4 ComputerName="FIN-WS-023" RemoteAddressIP4="203.0.113.77"
| table([@timestamp, ComputerName, ContextBaseFileName, ContextProcessId, LocalAddressIP4, LocalPort, RemoteAddressIP4, RemotePort])
```

Falcon CQL - Correlate Process Lineage to IOC

```
#event_simpleName=ProcessRollup2 ComputerName="FIN-WS-023" TargetProcessId=9820
| join({#event_simpleName=NetworkConnectIP4 RemoteAddressIP4="203.0.113.77"},
      key=ContextProcessId, field=TargetProcessId,
      include=[RemoteAddressIP4, RemotePort, LocalPort])
| table([@timestamp, ImageFileName, CommandLine, ParentBaseFileName, TargetProcessId, RemoteAddressIP4, RemotePort])
```

Falcon CQL - Reconstruct Suspicious Parent/Child Chain

```
#event_simpleName=ProcessRollup2 ComputerName="FIN-WS-023"
| in(field="TargetProcessId", values=[7412,8356,9820])
| table([@timestamp, ImageFileName, ParentBaseFileName, CommandLine, TargetProcessId])
```

Falcon Verification Result Matrix

Validation	Result	Meaning
NetworkConnectIP4 to 203.0.113.77	12 events	Falcon independently observes all twelve endpoint connections to the IOC.
ProcessRollup2 -> NetworkConnectIP4 join	12 correlated rows	Every observed connection maps to healthsync.exe PID 9820.
Process lineage	WINWORD.EXE -> powershell.exe -> healthsync.exe	Execution chain is inconsistent with ordinary HealthSync-style software deployment.
Process location	C:\Users\mcarter\AppData\Roaming\HealthSync\healthsync.exe	User-writable execution location increases suspicion.
Normal telemetry during window	Present	Sensor visibility is functioning.
Sensor health	Online / event flow normal	Falcon evidence is contemporaneous and complete enough for correlation.
Cross-check with Wireshark	Matches timestamps and destination	Falcon and packet evidence independently agree.

FALCON CONCLUSION

Falcon provides process attribution that the firewall alert lacks: healthsync.exe, launched by hidden PowerShell from WINWORD.EXE, is responsible for all twelve connections to 203.0.113.77:443.

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4. Sysmon Validation - Network, DNS, File, and Registry Evidence

Sysmon is used as a second endpoint evidence source. In the configured lab policy, Event ID 1 records process creation, Event ID 3 network connections, Event ID 22 DNS queries, Event ID 11 file creation, and Event ID 13 registry value changes. Coverage is validated before these events are used as corroborating evidence.

Raw Sysmon Events During the Alert Window

Line	Sysmon Source	Raw Sysmon Log / Query Result	Analyst Read
01	1 - Process Create	2026-08-18 12:19:41.499Z Image=C:\Program Files\Microsoft Office\root\Office16\WINWORD.EXE ParentImage=C:\Windows\explorer.exe User=CORP\mcarter ProcessGuid={TP-7412} CommandLine=WINWORD.EXE C:\Users\mcarter\Downloads\QuarterlyReview.docm	Document process begins.
02	1 - Process Create	2026-08-18 12:19:43.111Z Image=C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe ParentImage=WINWORD.EXE User=CORP\mcarter ProcessGuid={TP-8356} CommandLine=powershell.exe -NoProfile -WindowStyle Hidden -EncodedCommand <redacted>	Suspicious PowerShell child of Office.
03	11 - File Create	2026-08-18 12:19:44.716Z Image=powershell.exe TargetFilename=C:\Users\mcarter\AppData\Roaming\HealthSync\healthsync.exe CreationUtcTime=2026-08-18 12:19:44.716Z	PowerShell writes the synthetic malicious executable.
04	1 - Process Create	2026-08-18 12:19:45.079Z Image=C:\Users\mcarter\AppData\Roaming\HealthSync\healthsync.exe ParentImage=powershell.exe User=CORP\mcarter ProcessGuid={TP-9820} CommandLine=healthsync.exe -background	Dropped process executes.
05	13 - Registry Value Set	2026-08-18 12:19:45.392Z Image=healthsync.exe ProcessGuid={TP-9820} TargetObject=HKCU\Software\Microsoft\Windows\CurrentVersion\Run\HealthSync Details=C:\Users\mcarter\AppData\Roaming\HealthSync\healthsync.exe -background	Run-key persistence is created.
06	22 - DNS Query	2026-08-18 12:19:59.881Z Image=healthsync.exe ProcessGuid={TP-9820} QueryName=sync-check.example.invalid QueryStatus=0 QueryResults=203.0.113.77	Process resolves the C2 domain.
07	3 - Network Connect	2026-08-18 12:20:02.003Z Image=healthsync.exe ProcessGuid={TP-9820} SourceIp=10.40.23.55 SourcePort=52110 DestinationIp=203.0.113.77 DestinationPort=443 Protocol=tcp Initiated=true	First process-linked C2 connection.
08	3 - Network Connect	2026-08-18 12:21:02.005Z Image=healthsync.exe ProcessGuid={TP-9820} SourceIp=10.40.23.55 SourcePort=52147 DestinationIp=203.0.113.77 DestinationPort=443 Protocol=tcp Initiated=true	Second callback.
09	3 - Network Connect	2026-08-18 12:22:02.010Z Image=healthsync.exe ProcessGuid={TP-9820} SourceIp=10.40.23.55 SourcePort=52184 DestinationIp=203.0.113.77 DestinationPort=443 Protocol=tcp Initiated=true	Third callback.
10	3 - Network Connect	2026-08-18 12:25:02.009Z Image=healthsync.exe ProcessGuid={TP-9820} SourceIp=10.40.23.55 SourcePort=52295 DestinationIp=203.0.113.77 DestinationPort=443 Protocol=tcp Initiated=true	Sixth callback sample.
11	3 - Network Connect	2026-08-18 12:29:02.011Z Image=healthsync.exe ProcessGuid={TP-9820} SourceIp=10.40.23.55 SourcePort=52443 DestinationIp=203.0.113.77 DestinationPort=443 Protocol=tcp Initiated=true	Tenth callback sample.
12	3 - Network Connect	2026-08-18 12:31:02.012Z Image=healthsync.exe ProcessGuid={TP-9820} SourceIp=10.40.23.55 SourcePort=52517 DestinationIp=203.0.113.77 DestinationPort=443 Protocol=tcp Initiated=true	Twelfth callback.
13	query_result	2026-08-18 12:35:44.100Z EventID=3 DestinationIp=203.0.113.77 Image=healthsync.exe ResultCount=12	Twelve Sysmon network connections to the IOC.
14	query_result	2026-08-18 12:35:48.422Z EventID=22 QueryName=sync-check.example.invalid Image=healthsync.exe ResultCount=1	DNS resolution is attributed to the same process.
15	query_result	2026-08-18 12:35:55.011Z EventID=11 TargetFilename contains \HealthSync\ ResultCount=1	File creation confirms host state change.
16	query_result	2026-08-18 12:35:59.334Z EventID=13 TargetObject contains CurrentVersion\Run\HealthSync ResultCount=1	Persistence registry change is present.

ENDPOINT CONCLUSION

Sysmon independently corroborates Falcon and Wireshark: the same suspicious process is created, establishes persistence, resolves the C2 domain, and initiates twelve network connections to 203.0.113.77:443.

Cross-source timeline

EVIDENCE TRACK

- CASE BRIEF
- ALERT + RAW EVIDENCE
- PACKET VERIFICATION
- ENDPOINT CORROBORATION
- CORRELATION + JUDGMENT
- DETECTION ENGINEERING
- DISPOSITION + COMMUNICATION

TRUE POSITIVE

5. Normalized Cross-Source Investigation Timeline

The normalized timeline places the SIEM, raw firewall, packet, Falcon, and Sysmon evidence on one clock. Unlike the false-positive case, the independent sources reinforce rather than contradict the alert narrative.

Line	Time (Z)	Evidence Source	Key Evidence	What It Proves
01	12:19:41	Falcon / Sysmon	WINWORD.EXE opens QuarterlyReview.docm	Execution context begins before beaconing.
02	12:19:43	Falcon / Sysmon	WINWORD.EXE -> hidden encoded PowerShell	Suspicious parent-child execution.
03	12:19:44	Sysmon EID 11	PowerShell creates AppData\Roaming\HealthSync\healthsync.exe	Synthetic malicious artifact is written.
04	12:19:45	Falcon / Sysmon	powershell.exe -> healthsync.exe PID 9820	Beaconing process begins.
05	12:19:45	Sysmon EID 13	HKCU Run\HealthSync points to healthsync.exe	Persistence is created.
06	12:19:59	Wireshark / Sysmon EID 22	sync-check.example.invalid -> 203.0.113.77	C2 domain resolution is independently visible.
07	12:20:02	Firewall raw	10.40.23.55 -> 203.0.113.77:443 allow established bytes>0	First allowed outbound session.
08	12:20:02	Wireshark	SYN -> SYN/ACK -> TLS Client Hello -> application data	Packet sequence proves a real encrypted session.
09	12:20:02	Falcon / Sysmon	healthsync.exe PID 9820 -> 203.0.113.77:443	Endpoint process attribution matches packet timing.
10	12:21-12:31	Firewall / Wireshark	Eleven additional sessions recur at ~60-second intervals	Network periodicity is confirmed end-to-end.
11	12:21-12:31	Falcon / Sysmon	healthsync.exe produces all subsequent IOC connections	Same process drives the complete beacon series.
12	12:32:10	Parser	Direction=outbound, action=allow, bytes preserved	Normalization is correct.
13	12:32:10	Asset enrichment	10.40.23.55 mapped to FIN-WS-023 as local/source asset	Entity attribution is correct.
14	12:32:14	SIEM correlation	12 sessions / avg interval 60s / outbound + allow + established	Rule conditions are satisfied.
15	12:32:14	SIEM alert	High: Possible C2 Beaconing	Alert narrative matches the underlying evidence.
16	12:34:50	Wireshark pivot	IOC filter -> 56 frames; outbound SYN count=12	Packet capture independently confirms the alert.
17	12:35:10	Falcon CQL	NetworkConnectIP4 IOC count=12	EDR independently confirms endpoint connections.
18	12:35:22	Falcon join	12 IOC connections -> healthsync.exe PID 9820	Process attribution is established.
19	12:35:44	Sysmon	EID 3 count=12 plus EID22/EID11/EID13 evidence	Second endpoint source confirms network and state change.
20	12:38:00	Disposition	TRUE POSITIVE - confirmed C2 with persistence	Incident containment and eradication begin.

Evidence Corroboration Matrix

Question	SIEM Alert Claims	Independent Evidence	Judgment
Who initiated the traffic?	FIN-WS-023	Firewall + Wireshark show 10.40.23.55 as source; Sysmon Initiated=true	Confirmed endpoint initiation.
Was traffic allowed/established?	Implied successful beacon	Firewall action=allow, state=established, non-zero sent/received bytes	Confirmed session establishment.
Did packets reach/leave endpoint?	Yes	Wireshark: 56 IOC frames and 12 outbound SYN/TLS sessions	Confirmed packet path.
Did Falcon see a connection?	Endpoint C2	12 NetworkConnectIP4 events to IOC	Confirmed EDR network evidence.
Which process generated traffic?	Unknown in SIEM	Falcon joins all 12 sessions to healthsync.exe PID 9820	Process attribution confirmed.
Did Sysmon agree?	Not part of SIEM alert	12 EID 3 connections, EID 22 DNS, EID 11 file, EID 13 Run key	Independent host corroboration.
Did host state change?	Unknown	New AppData executable + HKCU Run persistence	Malicious effect confirmed.

TRUE POSITIVE

Analyst judgment and evidence priority

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6. Clean Line-by-Line Analyst Interpretation and SOC Judgment

Line	Analyst Identification	Why It Matters / Next Pivot	Priority
01	SIEM detects 60-second periodicity	Periodicity is a strong behavioral signal when direction and session validity are also established.	High
02	Raw firewall logs show outbound initiation	The protected endpoint is the source, matching the C2 narrative.	Critical
03	Firewall action is allow	Connections were permitted rather than blocked attempts.	Critical
04	Firewall state=established and bytes>0	Real bidirectional application sessions occurred.	Critical
05	Parser preserves direction/action/bytes	The alert is not caused by a normalization defect.	High
06	Asset enrichment maps source IP to FIN-WS-023	Host attribution correctly identifies the initiator.	High
07	Wireshark capture is continuous with zero drops	Packet evidence is trustworthy for direct correlation.	High
08	Wireshark shows 12 outbound SYN's and TLS sessions	Packet capture independently proves recurring C2-like communication.	Critical
09	Wireshark SNI matches sync-check.example.invalid	The recurring sessions consistently target the same controlled C2 domain.	Critical
10	Falcon NetworkConnectIP4 returns 12 IOC events	Endpoint sensor directly corroborates the claimed destination.	Critical
11	Falcon process-network join maps all 12 to healthsync.exe	The network activity is tied to one unknown executable.	Critical
12	Falcon process lineage is WINWORD -> PowerShell -> healthsync	Suspicious parent-child execution supports post-exploitation context.	Critical
13	Sysmon EID 3 returns 12 matching connections	Independent OS telemetry confirms Falcon network attribution.	Critical
14	Sysmon EID 22 records IOC DNS	Domain resolution path is present and process-linked.	High
15	Sysmon EID 11 records healthsync.exe creation	A new executable appears just before C2 begins.	Critical
16	Sysmon EID 13 records Run-key persistence	Host state change demonstrates persistence, not network-only anomaly.	Critical
17	Firewall, Wireshark, Falcon, and Sysmon align	Independent sources converge on one malicious chain.	Critical
18	SOC disposition	Classify TRUE POSITIVE, contain host, block IOC, preserve evidence, and scope environment.	Critical

Evidence Priority

Priority	Evidence	Reason
1	Raw firewall lines 01-12	Original source records prove outbound, allowed, established sessions with non-zero bytes.
2	Wireshark capture + integrity metadata	Packet sequence independently proves 12 handshakes/TLS sessions to the same destination.
3	Falcon NetworkConnectIP4 + process join	Endpoint telemetry links the network activity to healthsync.exe and suspicious lineage.
4	Sysmon Event IDs 1, 3, and 22	Independent OS telemetry confirms execution, DNS, and network behavior.
5	Sysmon Event IDs 11 and 13	File creation and Run-key modification prove malicious state change/persistence.
6	SIEM parser/correlation output	Shows the detection logic accurately represented the underlying data and generated a defensible alert.

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7. Investigation Pivots and Detection Validation

ANALYST CONCLUSION

TRUE POSITIVE. Raw firewall evidence proves allowed outbound established sessions; Wireshark captures the same repeated TLS callbacks; Falcon attributes all twelve sessions to healthsync.exe; Sysmon independently records the same process, DNS, network, file creation, and Run-key persistence. The SIEM alert narrative is therefore corroborated by multiple independent evidence sources.

CrowdStrike Falcon CQL - Confirm Endpoint Connection to IOC

```
#event_simpleName=NetworkConnectIP4 ComputerName="FIN-WS-023" RemoteAddressIP4="203.0.113.77"  
| table([@timestamp, ComputerName, ContextBaseFileName, ContextProcessId, LocalAddressIP4, LocalPort, RemoteAddressIP4, RemotePort])
```

CrowdStrike Falcon CQL - Require Process/Network Correlation

```
#event_simpleName=ProcessRollup2 ComputerName="FIN-WS-023" TargetProcessId=9820  
| join({#event_simpleName=NetworkConnectIP4 RemoteAddressIP4="203.0.113.77",  
      key=ContextProcessId, field=TargetProcessId,  
      include=[RemoteAddressIP4, RemotePort, LocalPort])  
| table([@timestamp, ImageFileName, CommandLine, ParentBaseFileName, RemoteAddressIP4, RemotePort])
```

Wireshark

```
ip.addr == 203.0.113.77  
ip.src == 10.40.23.55 && ip.dst == 203.0.113.77  
tcp.flags.syn == 1 && tcp.flags.ack == 0 && ip.src == 10.40.23.55 && tcp.dstport == 443  
tls.handshake.type == 1 && ip.dst == 203.0.113.77  
tls.handshake.extensions_server_name == "sync-check.example.invalid"
```

Sysmon Targeted Review

```
# Review the Sysmon Operational log over the exact alert window.  
# Pivots: Event ID 1 healthsync.exe / suspicious parent; Event ID 3 DestinationIp=203.0.113.77;  
# Event ID 22 QueryName=sync-check.example.invalid; Event ID 11 HealthSync file creation;  
# Event ID 13 CurrentVersion\Run\HealthSync persistence value.
```

Validated Correlation Logic - Design Requirements

Control	Required Condition	Why It Matters
Direction	normalized direction=outbound	Prevents inbound scans from being mislabeled as endpoint C2.
Action / disposition	allow/accept/success	Distinguishes established activity from blocked attempts.
Session establishment	established or meaningful byte evidence	Avoids treating isolated SYN attempts as completed C2 sessions.
Host attribution	source/local initiator semantics	Ensures the alert entity is the device that originated the traffic.
Periodicity	repeated sessions within a defined window	Detects automated callback behavior.
Destination context	external + rare/unapproved destination	Reduces noise from routine enterprise services.
Process corroboration	process-linked Falcon connection when endpoint telemetry is available	Raises confidence from network anomaly to endpoint-attributed C2.
State-change context	file/persistence evidence raises severity	Helps separate network-only anomalies from active compromise.

Detection logic and regression testing

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Conceptual Detection Logic and Regression Test Suite

Conceptual Validated Rule

// Field names vary by firewall parser. This is detection logic, not a vendor-specific parser contract.
network.direction = "outbound"
AND event.action IN ("allow","accept","success")
AND destination.port IN (80,443)
AND destination.ip is external
AND session.established = true
AND session.count >= 8 within 15m
AND periodicity_score >= threshold
AND destination not in approved-service baseline
// Increase confidence when a process-linked endpoint connection and suspicious lineage/state change also exist.

Regression Test Suite for the Analytic

Test Case	Input	Expected Result
A - This true positive	12 allowed outbound established TLS sessions at 60-second intervals + process-linked Falcon/Sysmon evidence	C2 rule fires High/Critical and is eligible for rapid containment.
B - False-positive counterpart	12 inbound denied SYN's at 60-second intervals	C2 rule does NOT fire; optional inbound-scan rule may fire.
C - Blocked outbound attempts	12 outbound denies with zero session bytes	C2 rule suppressed or downgraded; investigate policy separately.
D - Approved browser traffic	Periodic HTTPS from signed browser to approved service	C2 rule suppressed by reputation/baseline/process context.
E - Endpoint corroboration missing	Network source claims outbound C2 but Falcon/Sysmon process evidence is absent	Keep as suspicious / validation required rather than calling confirmed compromise.
F - Parser field loss	Direction or action becomes null during normalization	Data-quality test fails closed; do not generate a high-confidence C2 narrative.

Threat-Hunting Expansion After Confirmation

Pivot	Hunt Objective
Hash / file path	Find healthsync.exe or equivalent artifact on other endpoints.
Remote IP / domain	Identify every host that contacted 203.0.113.77 or resolved sync-check.example.invalid.
Parent-child lineage	Search for Office -> PowerShell -> user-space executable chains.
Run-key value	Find matching HealthSync persistence values across user profiles.
Process command line	Search for healthsync.exe --background and associated synthetic indicators.
User identity	Review mcarter authentication, downloads, mailbox, and lateral activity around initial execution.

Disposition and response plan

EVIDENCE TRACK

- CASE BRIEF
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8. True Positive Disposition, Incident Response, and Portfolio Explanation

SOC Disposition Record

Field	Final Value
Classification	TRUE POSITIVE
Alert	Possible C2 Beaconing - Repetitive HTTPS Sessions
Affected entity	FIN-WS-023 / 10.40.23.55
Confirmed IOC	203.0.113.77:443 / sync-check.example.invalid
Actual network behavior	Twelve outbound TCP/TLS sessions initiated by FIN-WS-023, allowed and established, with non-zero bidirectional data.
Wireshark	56 matching IOC frames; 12 outbound SYNs; 12 TLS Client Hello sessions; matching SNI and periodic timing.
CrowdStrike Falcon	12 NetworkConnectIP4 events to IOC; all correlate to healthsync.exe PID 9820; suspicious WINWORD -> PowerShell -> healthsync lineage.
Sysmon	12 Event ID 3 connections; Event ID 22 DNS; Event ID 11 file creation; Event ID 13 Run-key persistence; matching ProcessGuid.
Root cause	Synthetic adversary-emulation artifact healthsync.exe executed from AppData, established persistence, and beacons to controlled C2 infrastructure.
Host containment	Required.
Confidence	High - raw source, packets, EDR, and independent Sysmon evidence converge on the same chain.

Immediate Incident Response Plan

Phase	Action	Expected Result
1. Preserve evidence	Save SIEM alert, raw firewall events, parser output, PCAP/hash, Falcon query exports, Sysmon EVTX, process/hash/path, and registry evidence.	Creates a reproducible incident record.
2. Isolate endpoint	Use approved Falcon containment workflow for FIN-WS-023 after evidence requirements are met.	Stops further C2 and lateral activity.
3. Stop/quarantine artifact	Terminate healthsync.exe and quarantine the synthetic malicious artifact through approved EDR workflow.	Stops execution and preserves the artifact for analysis.
4. Remove persistence	Remove the HealthSync Run value after evidence capture and verify it does not recreate.	Prevents relaunch at user login.
5. Block infrastructure	Block 203.0.113.77 and sync-check.example.invalid at approved network/security controls.	Stops known callback path.
6. Secure identity	Reset/review the affected account and revoke sessions if the investigation shows credential exposure.	Prevents account reuse.
7. Hunt environment	Search hash, file path, process lineage, Run key, destination, domain, and user activity across the fleet.	Scopes additional compromise.
8. Recover and validate	Restore/clean endpoint as required, re-enable network access only after validation, and monitor for recurrence.	Provides evidence-based recovery.

Do Not Declare the Incident Resolved Until

Criterion	Required Evidence
C2 stopped	No new firewall, Wireshark, Falcon, or Sysmon connections to the IOC after containment.
Malicious execution removed	healthsync.exe is absent/quarantined and no equivalent process lineage recurs.
Persistence removed	HealthSync Run value and related autorun artifacts are absent.
Identity scoped	Affected user/session activity is reviewed and credentials are secured if exposure is suspected.
Fleet scope clean	Enterprise hunt finds no unexplained additional hosts or persistence artifacts.
Telemetry healthy	Falcon, Sysmon, network, and packet visibility remain operational after recovery.
Detection validated	True-positive replay fires; the false-positive inbound-deny regression case does not.

Professional case explanation

EVIDENCE TRACK

- CASE BRIEF
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Clean Professional Explanation

The SIEM raised a high-severity C2 beaconing alert because it observed twelve sessions involving FIN-WS-023 and 203.0.113.77 at almost exactly 60-second intervals. Unlike the false-positive version of this scenario, the normalized event retained the fields needed to support the narrative: direction was outbound, action was allow, TCP state was established, and every session contained non-zero sent and received bytes.

Review of the original firewall records therefore strengthened the alert rather than weakening it. FIN-WS-023 was consistently the traffic initiator. Wireshark then provided packet-level confirmation on the active endpoint interface. The capture showed DNS resolution of sync-check.example.invalid to 203.0.113.77, twelve outbound SYNs, completed TCP handshakes, twelve TLS Client Hello packets carrying the same SNI, encrypted application data in both directions, and nearly exact 60-second callback timing.

CrowdStrike Falcon added the process attribution that the network sources could not provide. NetworkConnectIP4 events showed twelve connections to 203.0.113.77:443, all generated by healthsync.exe PID 9820 from a user-writable AppData path. ProcessRollup2 placed that executable in a suspicious chain: WINWORD.EXE launched hidden encoded PowerShell, which launched healthsync.exe. This made the network behavior part of an execution story rather than an isolated IOC match.

Sysmon independently reproduced the same host narrative. Event ID 1 recorded the process chain, Event ID 11 recorded creation of healthsync.exe, Event ID 13 recorded a HealthSync Run-key value, Event ID 22 recorded the DNS query for the synthetic C2 domain, and Event ID 3 recorded the recurring network connections. The timestamps and process identity aligned with Falcon and Wireshark.

The incident is therefore classified as a true positive with high confidence. The decision is not based solely on periodicity, an external IP, or the SIEM severity. It is based on convergence: raw network logs establish outbound allowed sessions; Wireshark proves the packet sequence; Falcon identifies the responsible process and lineage; Sysmon confirms execution, DNS, network activity, file creation, and persistence. Containment, evidence preservation, indicator blocking, persistence removal, identity review, and enterprise scoping are justified by that combined evidence.

TRUE-POSITIVE STANDARD

A strong true-positive decision is the inverse of the false-positive investigation: the SIEM alert makes a specific claim, and independent raw, packet, EDR, and operating-system telemetry all support the same claim while also showing a malicious endpoint effect.

Discussion points and technical notes

EVIDENCE TRACK

- CASE BRIEF
- ALERT + RAW EVIDENCE
- PACKET VERIFICATION
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9. Portfolio Discussion Points and Technical Accuracy Notes

Portfolio Discussion Points

Question	Professional Answer Focus
Why was this a true positive?	The SIEM claimed outbound C2, and raw firewall records, Wireshark, Falcon, and Sysmon independently confirmed the same endpoint, destination, timing, and network direction. Falcon and Sysmon also tied the traffic to a suspicious process and persistence changes.
What did Wireshark prove?	It proved actual packet transmission: DNS resolution, outbound SYNs, completed TCP/TLS sessions, repeated SNI, bidirectional encrypted application data, and 60-second timing.
What did Falcon add?	Falcon linked all twelve connections to healthsync.exe PID 9820 and reconstructed the WINWORD -> PowerShell -> healthsync process lineage.
What did Sysmon add?	Sysmon independently recorded process creation, file creation, Run-key persistence, C2 DNS, and twelve initiated network connections.
Why is this stronger than an IOC match?	An IOC match alone can be incidental. Here the IOC is tied to execution, process lineage, state change, periodic sessions, and multiple independent sensors.
What was the immediate response?	Preserve evidence, contain the endpoint, stop/quarantine the artifact, remove persistence, block the IOC/domain, secure the identity, and hunt enterprise-wide.
How does this pair with the false-positive document?	The two documents use the same SIEM C2 hypothesis. In the false positive, raw/packet/endpoint evidence contradicts the alert. In this true positive, those sources converge and prove compromise.
What does this demonstrate?	SOC triage, packet analysis, EDR hunting, Sysmon validation, process attribution, cross-source correlation, true-positive decision quality, incident response, and detection engineering.

Technical Accuracy Notes

CrowdStrike Falcon Next-Gen SIEM supports CQL searching, and the supplied false-positive source document uses ProcessRollup2 for process events and NetworkConnectIP4 for process-linked IPv4 network activity. Exact fields available in a production tenant depend on licensing, sensor/platform, repository, and ingestion configuration.

Microsoft Sysmon event coverage is configuration-dependent. This synthetic lab explicitly assumes the deployed Sysmon configuration records Event ID 1 Process Create, Event ID 3 Network Connection, Event ID 22 DNS Query, Event ID 11 FileCreate, and Event ID 13 Registry Value Set activity used in this investigation.

The firewall vendor, parser, domain, IP addresses, executable, hash, and all raw records are synthetic. They are designed to demonstrate evidence correlation and incident reasoning, not to represent a real compromise or provide offensive instructions.

Source Transformation

Source Element	True-Positive Transformation
Supplied false-positive C2 alert	Preserved as the same core alert hypothesis for a direct portfolio comparison.
Raw firewall evidence	Changed from inbound/denied/zero-byte probes to outbound/allowed/established sessions with data transfer.
Wireshark	Changed from 0 IOC frames to repeated SYN/TLS/application-data sessions matching the alert.
CrowdStrike Falcon	Changed from 0 IOC events to 12 NetworkConnectIP4 events tied to healthsync.exe and suspicious process lineage.
Sysmon	Changed from no network/DNS/state-change evidence to matching process, DNS, network, file-create, and Run-key events.
Disposition	Changed from FALSE POSITIVE + parser/rule remediation to TRUE POSITIVE + containment, eradication, scope, and recovery validation.

FINAL PORTFOLIO STATEMENT - A true positive is not simply an alert that looks malicious. Here, the SIEM asserted a specific outbound C2 behavior, and the original network logs, packet capture, CrowdStrike endpoint telemetry, and Sysmon host evidence independently reconstructed the same malicious chain from execution and persistence through periodic encrypted communication. That convergence justifies containment and incident response.